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Survey of fake news detection using machine intelligence approach

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ABSTRACT

With the extensive spreading of all information through digital platforms, it is of maximal importance that each people get to differentiate between them. Fake news is a vast problem in our society we cannot predict which news is fake or real without having knowledge or proof of that particular news. This has become a supreme problem, so we decided to create a solution to this problem. Thus, we built a small model which helps in detecting fake news, where we are dealing with some articles which have been collected from the internet. We have labeled each of them as either fake or true. We have trained our dataset using these articles and have used different machine learning algorithms like Passive Aggressive Classifier, Naïve Bayes, Logistic Regression, Decision Tree, Long short term memory (LSTM), and Bidirectional Encoder Representations from Transformers (BERT) to compare the results. Our experimental result has achieved 99.6% accuracy from Decision Tree algorithm and obtained 99.8% recall from LSTM for detection of fake news. Passive Aggressive Classifier performs excellent on a large data set.

1. Introduction

With the advent of globalization and the rapid development of online platforms (including Facebook and Twitter), a good approach to information exchange has opened up that has never been ever seen in human history [1,2]. The propagation of false news also has a tremendous impact on the rest of the world. Fake news is also propagated via social media platforms such as Facebook and Twitter [3,4]. Our empowerment to create judgments is largely determined by the knowledge we absorb; our viewpoint is influenced by the information we consume. There is mounting evidence that people have responded irrationally to news that afterward proved to be false [5,6]. Our ability to make decisions is primarily impacted by the information we ingest; our perspective is influenced by the events we intake. People have reacted unreasonably to news that later turned out to be untrue, according to accumulating evidence.

This comprehensive machine learning (ML) based research article for identifying false news is concerned with both fake and genuine news [7]. Using the sklearn module, free python language-based ML library and term frequency inverse document frequency (TF-IDF) vectorizer, we can say about a token in our dataset. Then, we initialize ML models and fit the token. Here we have considered different ML models Passive Aggressive Classifier, Naïve Bayes algorithm, Logistic Regression, Decision Tree, LSTM and BERT.

In the end, the accuracy score and the confusion matrix tell us how well our model fares. This provides us an approximate value of our model, indicating if it is performing properly or not. After that, we can accept user input and determine whether it is phony or real [8,9].

This provides us an approximate value of our model, indicating if it is performing properly or not. After that, we can accept user input and determine whether it is phony or real [10]. Our remaining paper is organized in the form of the different types of

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algorithms those are explained in the related work section. The problem regarding the fake news is noted early. The solution to this problem and the algorithm used is presented in the evaluation section, then there are the final results written in the result and discussion section. Finally, conclusion is presented.

2. Related work

This section deals with the basics of the existing algorithms.

2.1. Passive aggressive classifier

Passive aggressive algorithms [11,12] are online learning algorithms similar with perceptron algorithm. When the classification's conclusion is achieved, then the algorithm stays passive for the right outcome but becomes aggressive when a mistake occurs and it updates and modifies. Its goal is to produce updates that fix the loss while producing little change in the weight vector's norm [13,14].

In contrast to batch learning, where the full training dataset is used at once, online machine learning algorithms take the input data in sequential order and update the machine learning model step by step. This is highly beneficial in circumstances where there is a large volume of data and training the full dataset is computationally impossible due to the sheer bulk of the data [15,16].

2.2. Naïve Bayes

A Naïve Bayes classifier is an algorithm that classifies entities using Bayes' theorem. Naïve Bayes classifiers are based on the assumption of substantial, or naive, independence between data point properties. Because they are simple to build, these classifiers are commonly employed in machine learning [17,18].

It is a straightforward method for building classifiers: models that give class labels to issue cases, which are expressed as vectors of attribute values, and the class labels are chosen from a limited set. For training such classifiers, there is no one algorithm, but rather a variety of algorithms based on the same principle: Naive Bayes assume that the value of one feature is independent of the value of any other feature, given the class variable. For example, if the fruit is red, round, and around 10 cm in diameter, it is termed an apple. Regardless of any possible relationships between the color, roundness, and diameter data, a Naive Bayes classifier considers each of these features to contribute independently to the likelihood that this fruit is an apple. In many real-world applications, the method of maximum likelihood is used to estimate parameters for naive bayes models; in other words, the naive bayes model may be used without adopting Bayesian probabilities or employing any Bayesian procedures [19,20].

2.3. Logistic regression

A logistic regression model analyzes the connection between one or more existing independent factors to predict a dependent data variable. A logistic regression, for example, may have been used to predict whether such a parliamentary candidate would win or lose an election, or whether a high school student would be admitted or not to a specific institution. These binary outcomes enable easy selection between two options [9]. In general, logistic regression refers to binary logistic regression with binary attribute values, but it may also predict two additional types of target variables.

2.4. Decision tree

It is a versatile tool that can be used in a variety of situations. Both classification and regression issues may be solved with decision trees. The name implies that it uses a tree-like flowchart to display the predictions that come from a sequence of feature-based splits. It commences with a root node and finishes with leaves making a decision [21].

Decision tree is a supervised classifier that may be used for classification and regression issues; however, it is most commonly employed to solve classification problems. Internal nodes contain dataset attributes, branches represent decision rules, and each leaf node provides the conclusion in a tree-structured classifier.

The decision node and the leaf node are the two nodes of a decision tree. Leaf nodes are the result of those decisions and do not include any more branches, whereas Decision nodes are used to make any decision and have several branches.

2.5. LSTM (Long Short-Term Memory)

LSTM stands for long short-term memory networks, used in the field of Deep Learning. It is a variety of recurrent neural networks (RNNs) that are capable of learning long-term dependencies, especially in sequence prediction problems. LSTM [22] has feedback connections, i.e., it is capable of processing the entire sequence of data, apart from single data points such as images.

2.6. BERT (Bidirectional Encoder Representations from Transformers)

It is based on transformers. It is a deep learning model where each output is attached to every single input. It is a pretrained model. Here high computational resource is needed. One paper [23] had been created regarding fake news detection on social media

	A	B	C
21410	U.S., Nortl	GENEVA (F	REAL
21411	U.S., Nortl	GENEVA (F	REAL
21412	Headless t	COPENHA(	REAL
21413	North Kore	UNITED N.	REAL
21414	'Fully comi	BRUSSELS	REAL
21415	LexisNexis	LONDON (	REAL
21416	Minsk cult	MINSK (Re	REAL
21417	Vatican up	MOSCOW	REAL
21418	Indonesia	JAKARTA (I	REAL
21419	Donald Tr	Donald Tru	FAKE
21420	Drunk Bra	House Inte	FAKE
21421	Sheriff Da	On Friday,	FAKE
21422	Trump Is S	On Christn	FAKE
21423	Pope Fran	Pope Franc	FAKE
21424	Racist Ala	The numbe	FAKE

Fig. 1. Showing the dataset used.

with a data mining perspective. This paper has been made with a lot of surveys. Fake news characterizes by psychological and social theories. Another paper [24] made with deep learning techniques has used Support Vector Machine (SVM). It is not suitable for large data sets. It does not perform very well when the data set has more noise. Another paper [25] made using sentiment analysis. The sentiment analysis is a very difficult task due to sarcasm. The words or text data implied in a sarcastic sentence come with a different sense of meaning depending on the senders or situations. Sarcasm is remarking someone's opposite of what you want to say. Sentiment analysis using BERT [26] algorithm gives insight about dealing capability with extraction of sentiment from financial news also.

3. Evaluation

The social media sites are highly strong and beneficial for discussing different vital issues for the society. It needs to follow ethics also. It is our responsibility to maintain veracity of spreading correct news.

3.1. Experimental setup

We ought to install python 3.7 or above, not the most recent version, since, in the most recent version, there may be an issue with the Scikit-learn module name (Sklearn). It is python's most useful and stable machine learning library. Then, we imported other libraries necessary for the program. Then we enter the dataset into the program then train the model and test using the different kinds of algorithms of which the results have been shown in the following pages.

3.2. Dataset

The data that we utilized is shown in this graphic Fig. 1. Where the title, text, and label three columns indicate that we have divided the dataset into two subsets, and then we match the specific keyword with the train data, and if positive is greater than negative, the result is True.

The Fig. 1 is the partial snapshot of the whole data [27] which is used for the survey.

3.3. Flow chart

The data is first collected and labeled in this model. It was then put up for improved tokenization and results. After that, we tokenized our train and test data and eliminate stop words like (for, it, do, oneself, and many others) which is feature extraction. To estimate the value and train the classifier, we apply the algorithm. The idf-train value was then fitted into the classifier, and the second half of the data was divided to train our model. The result is then presented in the form of a confusion matrix, and the model's accuracy is calculated. Which is depicted in the diagram Fig. 2.

3.4. Graphical representation

This graphical depiction Fig. 3 was generated by us and is derived from the data in Table 1. Fig. 3 is the depiction of a comparison of all the algorithms. In this case, the passive-aggressive classifier has higher accuracy.

The precision of naive bayes is higher, the recall of the passive aggressive classifier is higher, and the F- Measure of the passive aggressive classifier is higher.

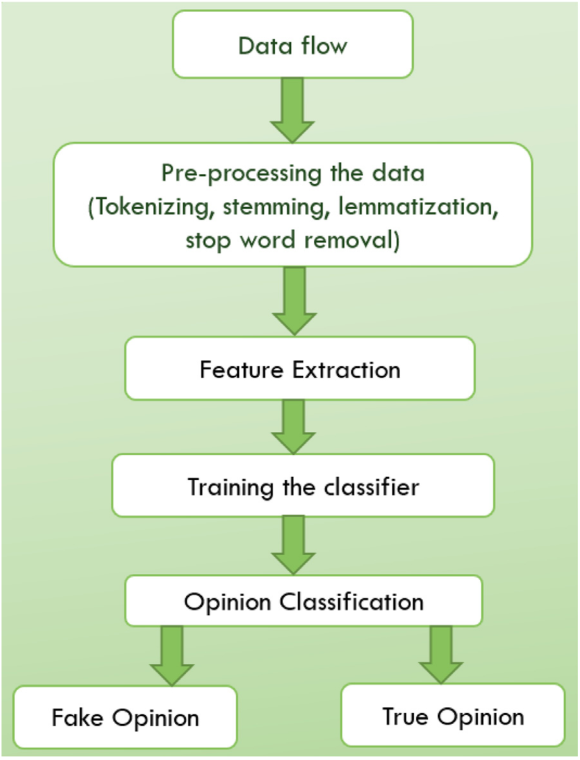

Fig. 2. Flow chart of the whole process undergoing.

Table 1
Result obtained from different algorithms.

S/N	Algorithm	Accuracy	Precision	Recall	F Measure
1	Passive Aggressive Classifier	99.520	99.656	99.421	99.538
2	Logistic Regression	98.594	99.049	98.242	98.644
3	Naïve Bayes	93.888	93.659	94.663	94.159
4	Decision Tree	99.609	99.529	99.721	99.625
5	LSTM	97.992	96.458	99.807	98.104
6	BERT	89.80	100	50	66.67

3.5. Important formulae

The various formulae which we have used in calculating Accuracy, Precision, Recall and F-measure are written in this section.

We have taken these into consideration for comparing the various algorithms like passive aggressive, naïve bayes, logistic regression, LSTM, BERT.

1. *Accuracy Calculation*

Accuracy is the quality of exactness. If we would like to draw conclusions from a given set of data, then we can take help of accuracy measurement.

In machine learning point of view, it will give the concept of the ratio between the number of correct predictions among total number of predictions. It is related to the observation error.

$$Accuracy = \frac{(Total\ number\ of\ correct\ predictions)}{Total\ number\ of\ prediction}$$

$$Accuracy = \frac{(T + TN)}{T + TN + FP + F}$$

T is true positive.

TN is true negative

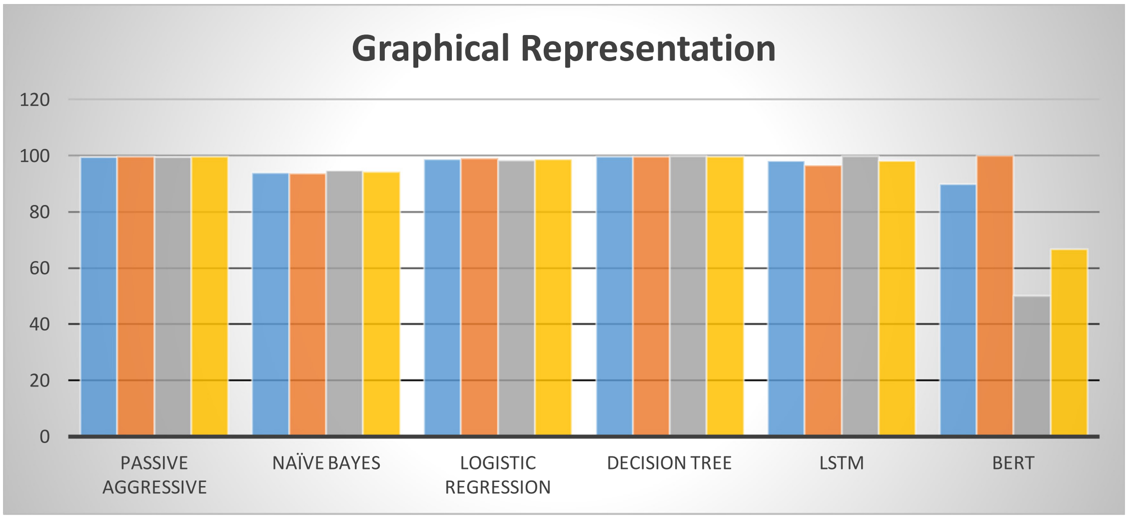

Fig. 3. Graphical representation depicting all the different methods.

FP is false positive
F is false negative
2. *Precision Calculation*

Precision measures how close the results are to one another. The number of positive class forecasts that really fall within the positive class is measured by precision. So, it predicts correct positive part in minority data class. Here we like to concentrate on false positive minimization.

$$Precision\ Ratio = \frac{T}{T+F}$$

T is the number of true positives and
F is the number of false positives.
3. *Recall Calculation*

Recall is the ability of a model to find all the relevant cases in a dataset. Recall measures how many correct class predictions were produced using all of the successful cases in the dataset. This metric quantifies correct positive predictions among all positive predictions. Minimization of false negative prediction is focused through recall parameter.

$$Recall\ Ratio = \frac{T}{T+F}$$

T is the number of true positives and
F is the number of false negatives.
4. *F-measure Calculation*

F-measure is the harmonic mean of precision and recall. F-Measure offers a single score that reconciles the issues of memory and accuracy in one number.

$$F - Measure = \frac{2*\ Precision * Recall}{Precision + Recall}$$

3.6. Confusion matrix

A confusion matrix is a summary of classification problem prediction outcomes. The number of rights and bad predictions is collected and split down by class using count values. The confusion matrix depicts the various ways in which our classification model becomes perplexed while making predictions. It reveals not only the sorts of errors produced by the classifier but also the number of errors made by the classifier.

The confusion matrix for each of the four algorithms is illustrated in the following figures Figs. 4, 5, 6, 7, 8, 9 with different values for True Positive, False Negative, False Positive, and True Negative.

4. Results and discussion

We have trained this model every time, this false news detection algorithm always returns an accurate score. When we train our model with a larger set of data, we get approximate and the finest data-testing outcome. It is simple to calculate the approximate

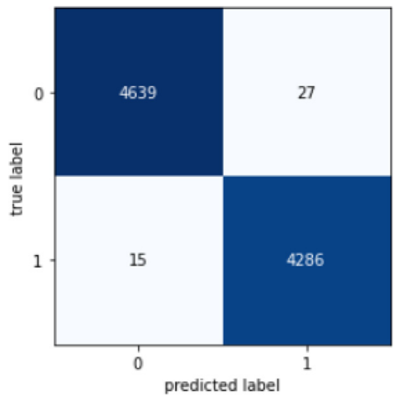

Fig. 4. Confusion matrix of passive aggressive classifier.

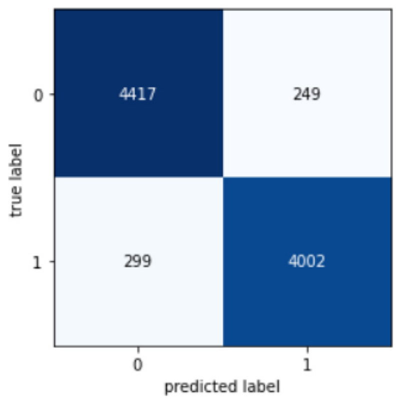

Fig. 5. Confusion matrix of naïve bayes.

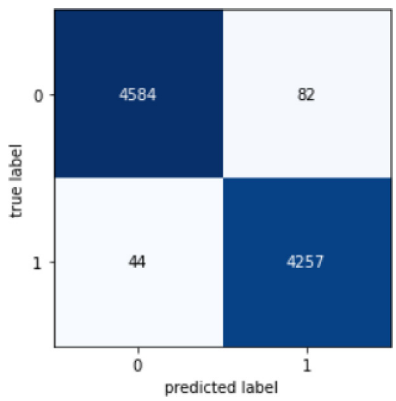

Fig. 6. Confusion matrix of logistic regression.

value using the machine learning idea of calculating the mean value of the generated vector and comparing it to our train datasets to see whether the number of positive outcomes is more than the news is true or false.

According to accuracy, passive aggressive classifier has 99.520, logistic regression has 98.594, naïve bayes has 93.888, decision tree has 99.609, LSTM has 97.992 and BERT has 89.80. Among all these Decision Tree has the highest value.

In accordance to precision, passive aggressive classifier has 99.656, logistic regression has 99.049, naïve bayes has 93.659, decision tree has 99.529, LSTM has 96.458 and BERT has 100. Among all these BERT has the highest value (in small datasets) [28].

In accord with recall, passive aggressive classifier has 99.421, logistic regression has 98.242, naïve bayes has 94.663, decision tree has 99.721, LSTM has 99.807 and BERT has 50. Among all these LSTM has the highest value.

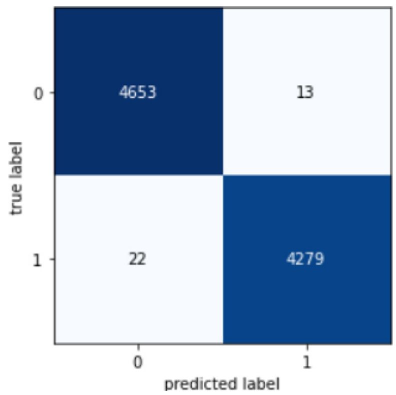

Fig. 7. Confusion matrix of decision tree.

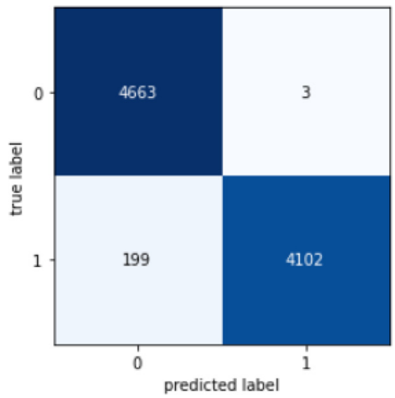

Fig. 8. Confusion matrix of LSTM.

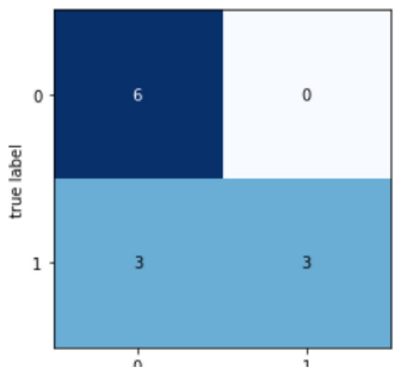

Fig. 9. Confusion matrix of BERT.

In consonance with F-measure, passive aggressive classifier has 99.538, logistic regression has 98.644, naïve bayes has 94.159, decision tree has 99.625, LSTM has 98.104 and BERT has 66.67. Among all these Logistic Regression has the highest value.

After looking into all the parameters, we can conclude that passive aggressive classifier is providing the best result on an average. For better analysis of all the algorithms Table 1 is shown.

5. Conclusions

We categorized the news in a word set and discovered the proper key terms to create an appropriate model. We have used different ML algorithms to train our model. Confusion matrix is depicted for measuring performance to make correct decisions concerning articles labeled as real or fake.

There are several outstanding challenges in the identification of fake news that researchers must address. Identifying essential aspects involved in the distribution of news, for example, can help to minimize the spread of fake news. In this article we try to establish comparative analysis of different ML algorithms for the fake news detection problem. Our research gives insight about ML application for the specified problem, which helps our society and new researchers in this field. In the future, we will combine this model with a new algorithm to produce a much broader outcome and eliminate any misunderstandings.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

We downloaded the data from Kaggle and [ISDDC 2017] ISOT [27]. GitHub provided us with the procedures for various areas of the code, which we appreciate. We appreciated it because we were able to learn about all of these current events and placed them into being used in this project.

BERT model is very huge because of the training structure. Due to its size and the number of weights that need to be updated, training is sluggish. So, we have used a short data set for it which we have uploaded on GitHub for Ref. [28].

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